

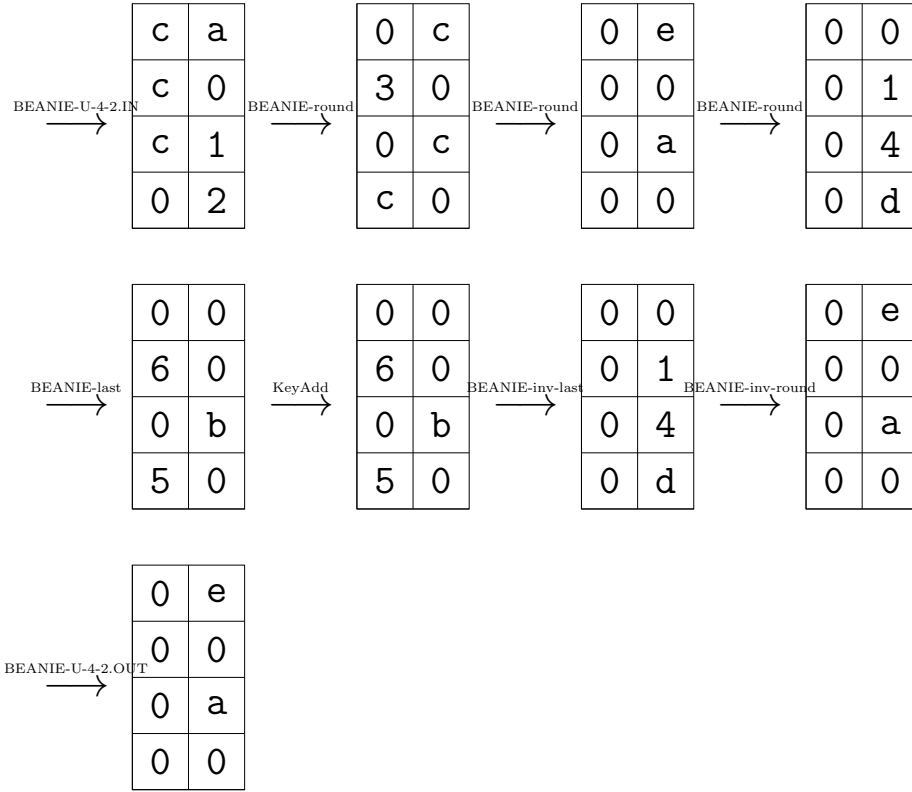
Bitwise differential trail through **BEANIE-U-4-2** found by SAT

CiVerLy

July 16, 2026

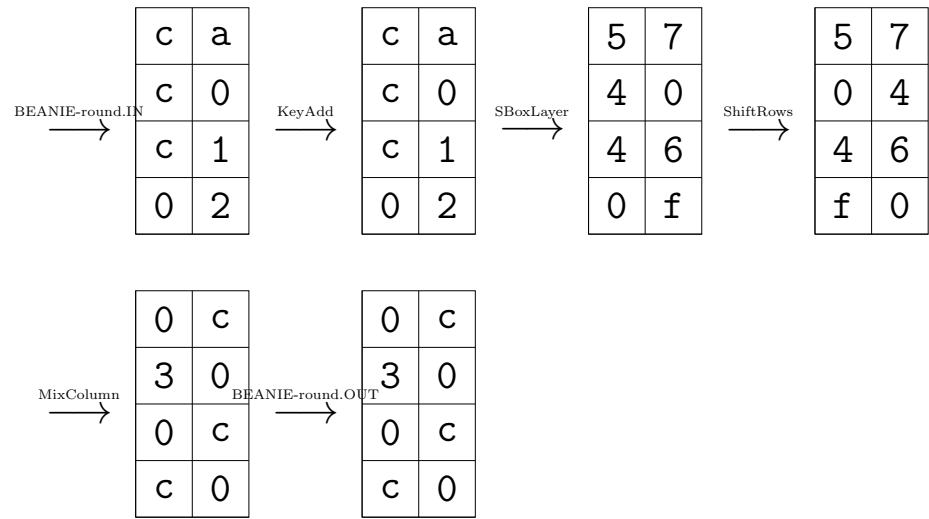
1 BEANIE-U-4-2

Maximal differential probability: 2^{-40}



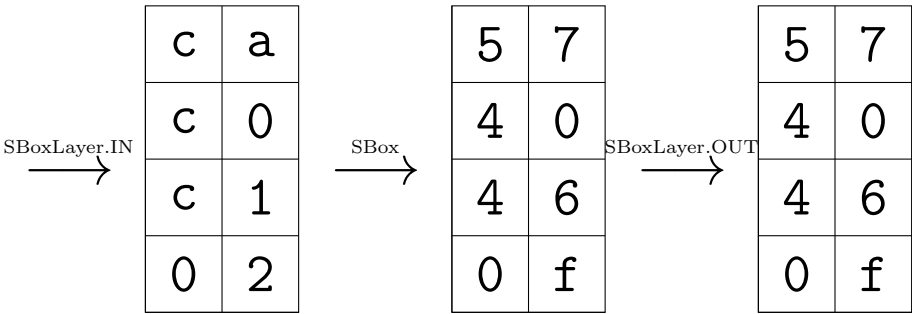
2 BEANIE-round

Maximal differential probability: $2^{-12.0}$



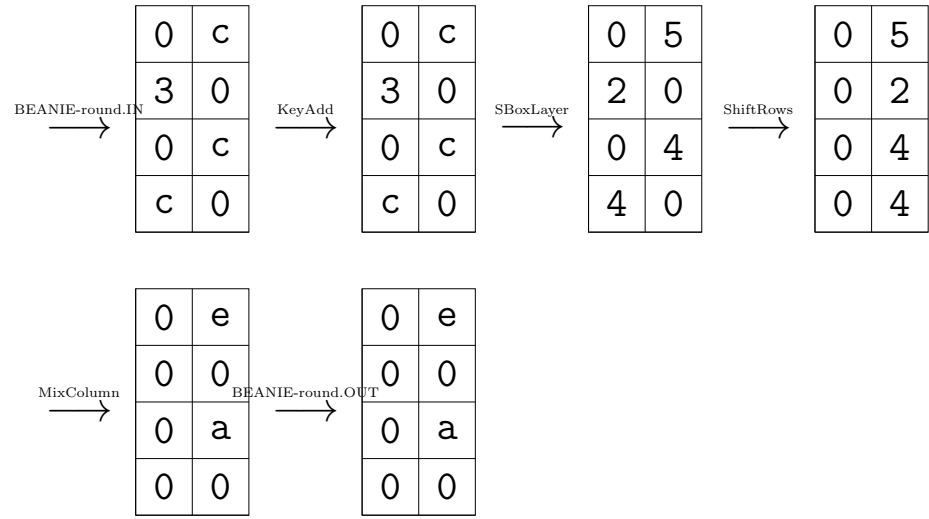
3 SBoxLayer

Maximal differential probability: $2^{-12.0}$



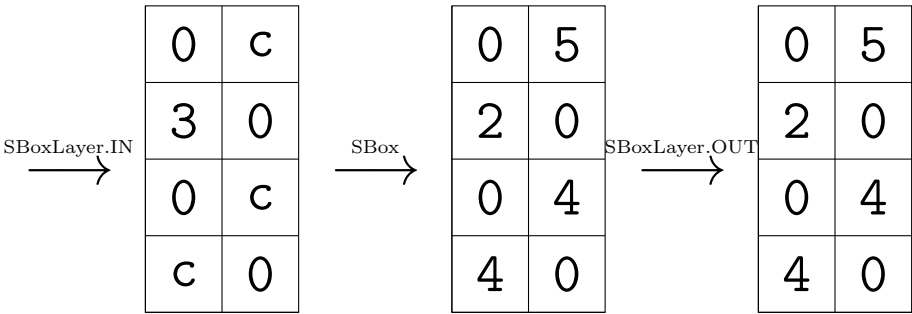
4 BEANIE-round

Maximal differential probability: $2^{-8.0}$



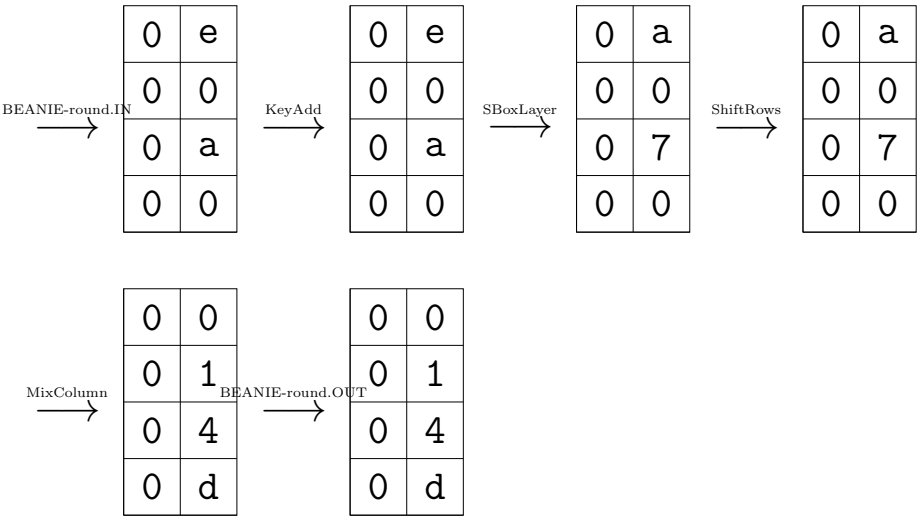
5 SBoxLayer

Maximal differential probability: $2^{-8.0}$



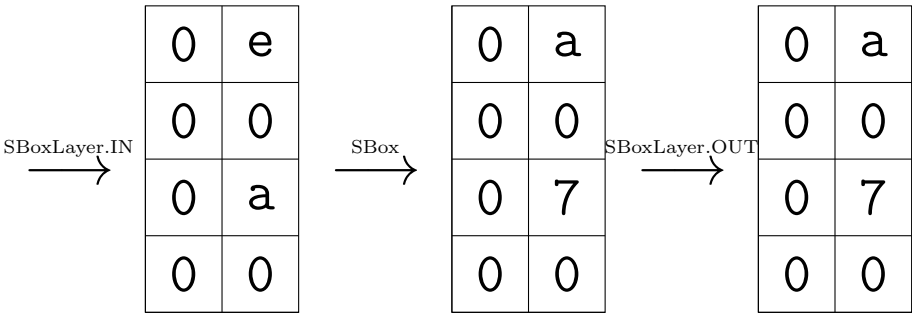
6 BEANIE-round

Maximal differential probability: $2^{-4.0}$



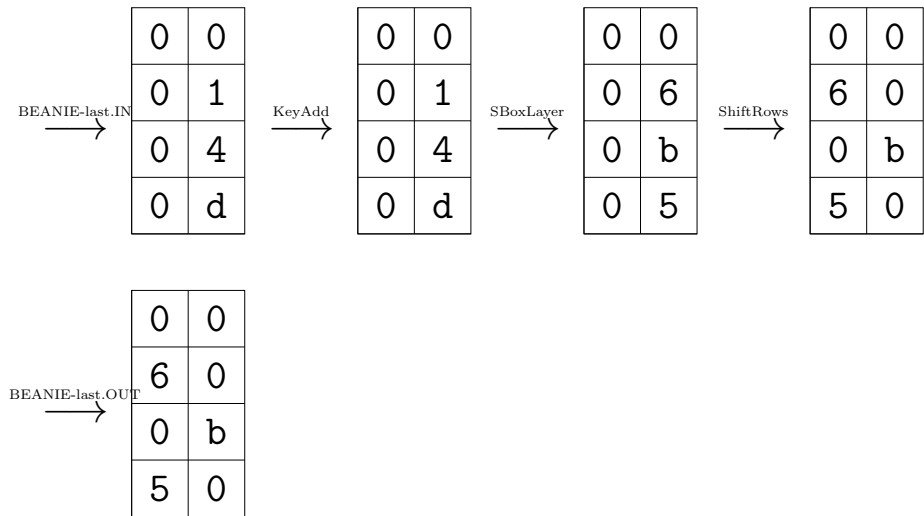
7 SBoxLayer

Maximal differential probability: $2^{-4.0}$



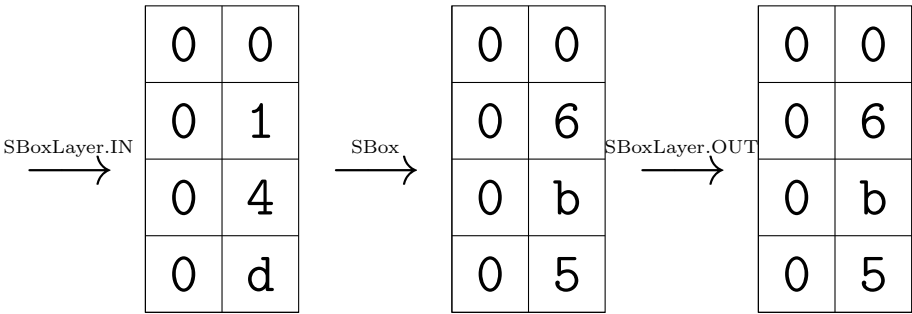
8 BEANIE-last

Maximal differential probability: $2^{-6.0}$



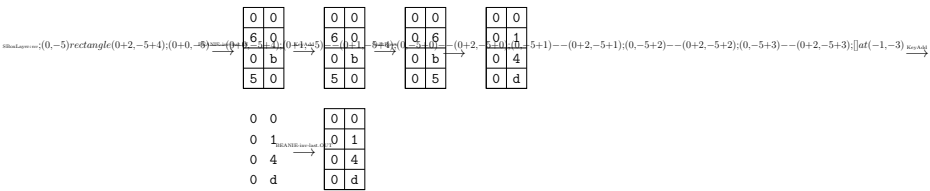
9 SBoxLayer

Maximal differential probability: $2^{-6.0}$



10 BEANIE-inv-last

Maximal differential probability: $2^{-6.0}$

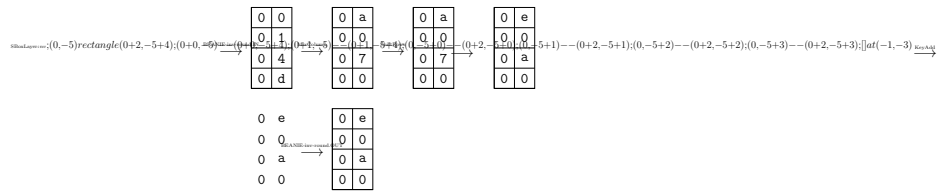


11 SBoxLayer_inv

Maximal differential probability: $2^{-6.0}$

12 BEANIE-inv-round

Maximal differential probability: $2^{-4.0}$



13 SBoxLayer_inv

Maximal differential probability: $2^{-4.0}$